SPATIAL REORGANIZATION OF ACTIN AND MICROTUBULES AND ITS ASSOCIATION WITH NUCLEAR MORPHOLOGY IN DRUG-TREATED MCF-7 CELLS

Nakulan (255HSBD017) | Pujasvi Jella (255HSBD018)

Aim

To compare nuclear morphology and the relative distribution of F-actin- and β-tubulin-associated fluorescence around nuclei in DMSO-, Cytochalasin B-, and Taxol-treated MCF-7 cells. We also assessed whether segmentation and region-of-interest (ROI) quality control changed the interpretation of these measurements.

Methodology

We analysed 144 fluorescence images from 48 fields across three BBBC021 plates (DMSO: 24 fields; Cytochalasin B: 12; Taxol: 12). Every field contained DAPI (nuclear DNA), phalloidin-labelled F-actin, and antibody-labelled β-tubulin channels.

Using Fiji/ImageJ, DAPI images were duplicated, thresholded with Otsu, converted to binary masks, filled to remove internal holes, and processed by Watershed to attempt separation of touching nuclei. Analyze Particles generated ROIs and nuclear area, circularity and solidity. Perinuclear fluorescence was measured in two bands outside each ROI (0–8 and 8–16 pixels). For each channel, the field-specific 10th-percentile intensity was subtracted from both ring means; the spatial ratio was corrected inner mean / corrected outer mean.

Morphology screening required area ≥100 px² and ≥16 px clearance from image borders. Strict fluorescence QC additionally required no overlap of either ring with other nuclear-mask pixels and positive corrected values in both rings and both channels. Three manually processed fields were visually checked using saved Fiji ROI overlays.

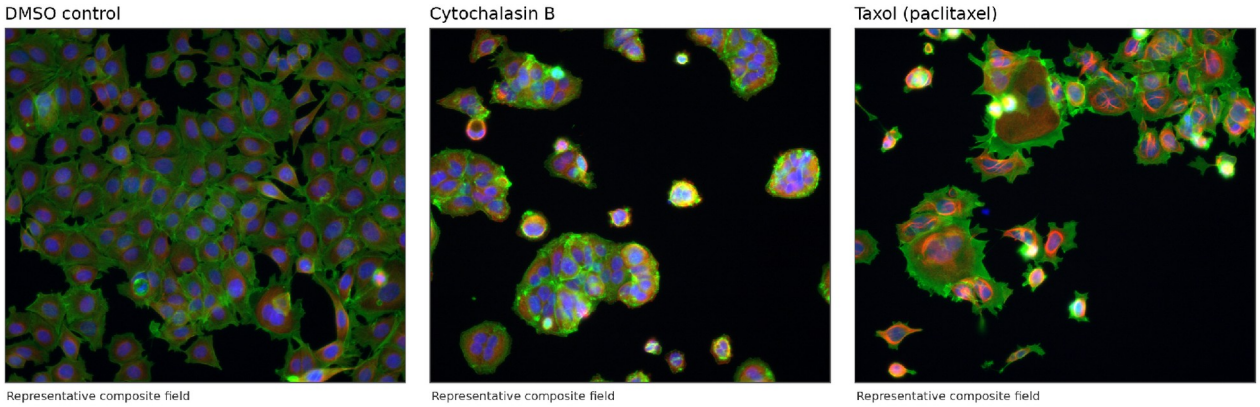


Figure 1. Representative three-channel composites used for comparison (left to right: DMSO, Cytochalasin B, Taxol).

Results

Across 48 fields, Fiji detected 8,435 nuclear objects. Of these, 6,484 passed size/edge screening and 1,967 passed strict perinuclear QC. Detection is not equivalent to confirming every object as an intact nucleus.

Condition	Detected objects	Morphology-screened	Strict ring-QC
DMSO	5,678	4,718	1,694
Cytochalasin B	1,497	1,132	139
Taxol	1,260	634	134

Circularity and solidity describe nuclear shape; inner/outer ratios describe relative fluorescence. Unequal strict-QC retention prompted relaxed-filter sensitivity analysis.

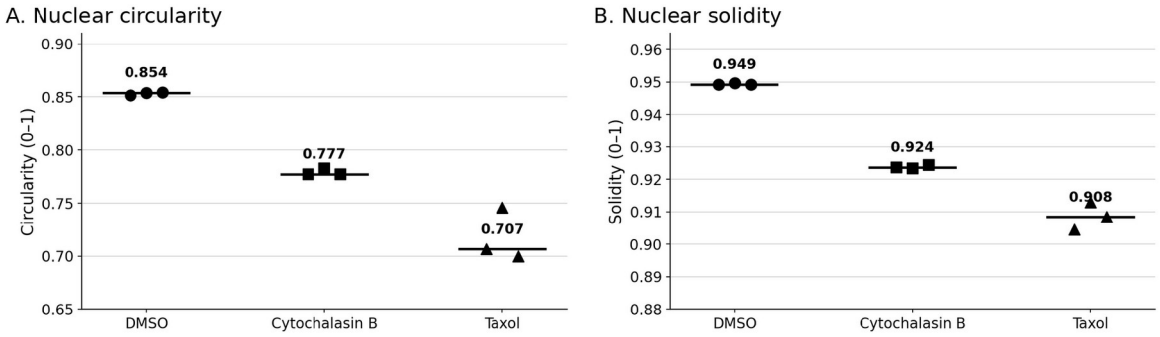


Figure 2. Plate-level medians (three plates per condition) after size and edge screening. Dots are plate summaries; horizontal marks indicate medians across plates.

Plate medians showed lower circularity after Cytochalasin B (0.777) and Taxol (0.707) than DMSO (0.854); solidity followed the same ordering (0.924, 0.908, 0.949, respectively). Fluorescence ratios, however, were strongly affected by ROI selection: for Cytochalasin B, the F-actin ratio fell from about 2.49 under strict QC to about 1.10 when overlap exclusion was relaxed. β-tubulin showed comparable sensitivity. These findings are descriptive, and do not establish drug efficacy, cell death or causation.

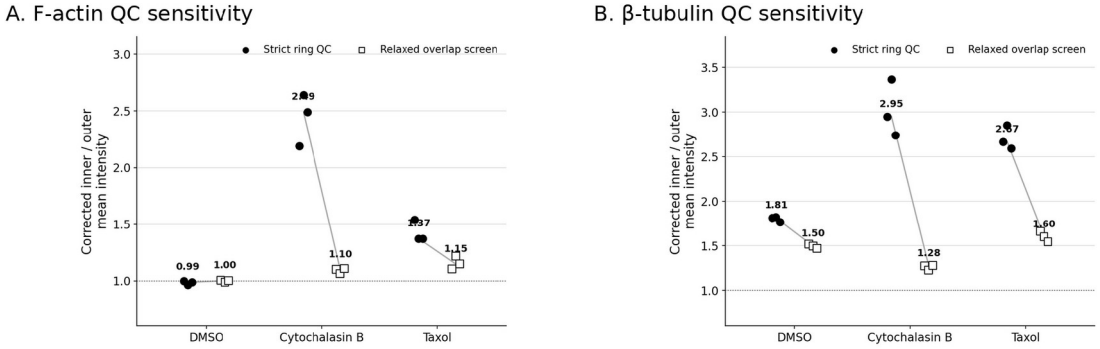


Figure 3. Ring fluorescence ratios under strict versus relaxed overlap rules. Markers are plate summaries; shifts show sensitivity to ROI selection.

Theoretical understanding gained

Biologically, we learned how each stain relates to the cytoskeleton: phalloidin labels filamentous actin (F-actin), while a β-tubulin antibody visualizes tubulin-associated structures built from α/β-tubulin subunits. Cytochalasin B interferes with actin assembly, whereas Taxol suppresses normal microtubule dynamics. Fluorescence shows the distribution of labelled material, but does not directly establish drug efficacy or cell death.

Analytically, we understood how Otsu thresholding identifies nuclear foreground, Fill Holes repairs internal mask gaps, and Watershed attempts to separate touching nuclei. The resulting ROIs enable area, circularity and solidity measurements. Measuring fluorescence within 0–8 px and 8–16 px bands adds spatial information that whole-image averages cannot provide.

The exercise also clarified the difference between nuclear shape, protein distribution and protein abundance. Combining these measurements helps examine cell architecture under different treatments, while interpreting the observed changes requires separating structural evidence from claims about cell function or survival.